Sampling Methods & Bias

ANSWER KEY

Identifying sampling methods

1 A school assigns each student a number from 1 to 800, then uses a random number generator to select 40 numbers, surveying those students. Identify this sampling method.

This is a simple random sample (SRS): every student has an equal chance of being selected, and every group of 40 students has an equal chance of being the chosen sample.

2 A researcher divides a school into grade levels (9th, 10th, 11th, 12th), then takes a simple random sample of 20 students from each grade. Identify this sampling method and explain why it might be preferred over an SRS of the whole school.

This is stratified random sampling. It might be preferred because it guarantees representation from every grade level, which is useful if opinions or characteristics vary meaningfully by grade — an SRS of the whole school could, by chance, under-represent or over-represent a particular grade.

Cluster sampling and systematic sampling

3 A researcher randomly selects 5 entire homeroom classes out of 40 in a school, and surveys every student in those 5 classes. Identify this sampling method, and contrast it with stratified sampling.

This is cluster sampling: entire naturally-occurring groups (homerooms) are randomly selected, and every member of the selected groups is surveyed. This differs from stratified sampling, where the researcher samples from every group, rather than sampling only a few whole groups and surveying everyone within them.

4 A quality-control inspector checks every 25th item coming off a production line. Identify this sampling method.

This is systematic sampling: individuals are selected at a fixed interval (every 25th item) from an ordered list or sequence, typically starting from a randomly chosen point.

Sources of bias

5 A survey about a new school policy is voluntarily filled out online, and only 12% of students respond. Identify the type of bias most likely present, and explain why the results may not represent the whole student body.

This is voluntary response bias: students with strong opinions (especially strongly negative or strongly positive ones) are more likely to take the time to respond, so the 12% who responded are not representative of the full student body's range of opinions — moderate or indifferent students are underrepresented.

6 A phone survey about internet usage habits is conducted using only landline phone numbers. Identify the type of bias this introduces.

This introduces undercoverage bias: households that only use cell phones (increasingly common, especially among younger people) are systematically excluded from the sampling frame, so the sample cannot represent the full population of interest.

Wording and response bias

7 A survey question reads: "Don't you agree that the wasteful new cafeteria policy should be reversed?" Explain why this question is problematic, and rewrite it in a neutral form.

The question is leading — the word "wasteful" and the phrasing "don't you agree" push respondents toward a particular answer, biasing the results. A neutral rewrite: "Do you support or oppose the new cafeteria policy?"

8 Explain why respondents might not answer a sensitive question (such as about illegal activity) truthfully in a face-to-face interview, and name this type of bias.

This is response bias: when asked sensitive questions directly by an interviewer, respondents may feel social pressure to give a more socially acceptable answer rather than the truthful one, especially without the anonymity that a written or online survey might provide.

Comparing sample size and bias

9 Explain why increasing the sample size does not fix a biased sampling method, using the voluntary response survey from earlier as an example.

Bias is a systematic flaw in how the sample was selected, not a matter of the sample being too small — collecting responses from a larger number of self-selected volunteers would still overrepresent people with strong opinions. Only fixing the sampling method itself (e.g. switching to a simple random sample of all students) removes the bias; a larger biased sample is still biased, just with a more precise estimate of the wrong quantity.

Designing a sampling plan

10 A city has 4 distinct neighborhoods with very different income levels, and the mayor wants to survey residents about a new tax proposal in a way that fairly represents each neighborhood. Recommend a sampling method, and describe how it would be carried out.

Stratified random sampling is recommended: divide the population into the 4 neighborhoods (strata), then take a simple random sample of residents from each neighborhood, proportional to (or intentionally balanced across) each neighborhood's population. This ensures every income group is represented, rather than relying on chance to include them proportionally as an SRS of the whole city might not.

Convenience sampling and its pitfalls

11 A researcher stands outside a gym and surveys the first 50 people who walk out about their exercise habits, intending to represent the whole city's exercise habits. Identify this sampling method and explain the flaw in generalizing its results to the whole city.

This is convenience sampling: the researcher selects whoever is easiest to reach rather than using any random selection process. The results cannot be generalized to the whole city because people at a gym are not representative of the general population — they are systematically more likely to exercise regularly than the average city resident, biasing the sample toward higher reported exercise habits.

Comparing sampling methods side by side

12 A national retailer wants to estimate average customer satisfaction across all 500 of its stores. Compare how a cluster sample (randomly selecting 20 whole stores and surveying all customers there) and a stratified sample (grouping stores by region, then sampling customers from every region) would each be carried out, and state one advantage of each approach.

Cluster sampling: randomly select 20 stores, then survey all customers at those 20 stores — cheaper and more logistically convenient since fieldwork is concentrated in fewer physical locations. Stratified sampling: divide the 500 stores into regions (strata), then sample customers from every region — guarantees geographic representation across all regions rather than relying on the 20 randomly chosen stores to happen to cover every region well.

Identifying the population and sample

13 A university wants to know the average commute time for all 30,000 enrolled students, and surveys a random sample of 400 students. Identify the population and the sample in this scenario.

Population: all 30,000 enrolled students at the university. Sample: the 400 students who were randomly selected and surveyed.

14 A national news outlet runs a text-in poll during a live broadcast, asking viewers to text "YES" or "NO" on a controversial policy question, and reports the results as representing "public opinion." Identify every source of bias you can find in this scenario, and explain why the reported percentages should not be trusted as an accurate reflection of the general public's views.

This scenario has at least two compounding sources of bias: (1) undercoverage — only viewers of that specific broadcast, who own a phone capable of texting and chose to watch that program, are even eligible to respond, excluding large portions of the public; (2) voluntary response bias — among those eligible, only viewers with strong opinions (often skewed toward one side) are motivated to actually text in. Together, these make the reported percentages reflect the opinions of a self-selected, unrepresentative subgroup, not the general public.

15 A hospital wants to estimate the average patient satisfaction score across its 12 different departments (e.g. cardiology, pediatrics, emergency), which are expected to have very different typical satisfaction levels. Describe, step by step, how you would design a stratified random sample to estimate overall patient satisfaction, and explain why a simple random sample of all patients might be less informative for comparing satisfaction across departments.

Step 1: treat each of the 12 departments as a separate stratum. Step 2: take an independent simple random sample of patients from within each department. Step 3: combine the department-level samples into the overall estimate (optionally weighting by each department's size). This guarantees every department is represented and allows department-by-department comparisons. A simple random sample of all patients, by contrast, might by chance draw very few patients from a smaller department, making it hard to estimate or compare that department's satisfaction reliably.

16 A researcher wants to estimate the average time it takes customers to be served at a chain of 200 coffee shops nationwide. Due to budget constraints, the researcher can only visit a small number of physical locations. Describe how a cluster sample could be carried out in this situation, and explain a key risk of cluster sampling that would not apply if the researcher instead used a stratified sample based on region.

A cluster sample would involve randomly selecting a small number of entire coffee shops (say, 15 out of 200) and measuring service time for every customer at those 15 locations. A key risk is that if certain regions or store types happen to differ meaningfully in typical service time, the randomly selected 15 shops could, purely by chance, fail to represent that variation well (e.g. all 15 could happen to be busy urban locations). A stratified sample based on region would guarantee representation from every region by design, rather than leaving it to chance which regions end up represented.